

HANEDA AP, Japan

WMO: 476710

Lat: 35.553N

Lon: 139.781E

Elev: 9

StdP: 101.21

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	1.9	-13.0	1.2	4.2	-11.1	1.5	4.6	13.2	8.3	12.2	7.7	5.3	330	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	33.0	26.1	31.9	25.6	30.9	25.4	26.9	30.5	26.4	29.9	25.9	29.4	7.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	29.3	25.2	20.3	28.7	24.9	20.0	28.5	84.8	31.2	82.4	30.0	80.2	29.5	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.2	10.1	-1.1	35.2	1.1	1.4	-1.9	36.1	-2.5	36.9	-3.1	37.7	-3.9	38.7		
(5)			-4.3	27.7	1.0	0.9	-5.0	28.4	-5.6	28.9	-6.2	29.4	-6.9	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	6.3	6.8	9.8	14.7	19.2	22.2	26.3	27.8	24.2	19.1	13.7	8.8
	DBStd	7.77	2.04	2.49	3.10	3.12	2.57	2.39	2.68	2.38	3.00	2.65	2.65	2.45
	HDD10.0	311	117	96	42	3	0	0	0	0	0	0	2	52
	HDD18.3	1552	373	324	263	114	21	2	0	0	1	21	140	294
	CDD10.0	2735	2	6	37	144	286	367	504	551	427	283	113	17
	CDD18.3	933	0	0	0	4	48	118	246	292	177	46	1	0
	CDH23.3	7256	0	0	0	6	103	457	2224	3107	1242	116	1	0
	CDH26.7	2256	0	0	0	0	7	66	702	1161	308	12	0	0
Wind	WSAvg	5.2	5.1	5.3	5.6	5.7	5.4	4.9	5.3	5.3	5.1	5.0	4.7	4.8
Precipitation	PrecAvg	1548	62	64	112	124	137	160	149	145	218	209	108	61
	PrecMax	2040	141	141	198	234	256	319	325	356	462	683	256	198
	PrecMin	1142	3	4	36	44	62	44	31	14	46	18	8	1
	PrecStd	219	40	36	39	51	50	58	92	95	104	147	60	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	18.1	20.8	24.1	27.8	30.2	34.0	34.9	32.2	28.1	22.8	19.1
		MCWB	11.6	13.3	14.3	17.3	19.3	23.5	25.6	26.5	24.9	22.0	17.3	14.2
	2%	DB	13.1	15.0	18.2	22.1	25.8	28.2	32.2	33.2	31.0	26.0	20.8	16.0
		MCWB	8.4	10.4	13.1	16.0	19.1	22.3	25.6	26.3	24.9	21.2	16.1	11.3
	5%	DB	11.2	12.2	16.9	21.0	24.2	27.0	31.2	32.2	29.8	24.2	19.1	14.1
		MCWB	6.4	7.8	12.5	15.9	18.5	21.9	25.5	25.8	24.5	19.9	14.9	9.7
	10%	DB	10.1	10.9	15.1	19.8	23.1	25.9	30.1	31.2	28.2	23.1	18.0	13.0
		MCWB	5.5	6.5	10.7	14.9	18.0	21.4	25.1	25.7	23.8	19.1	14.1	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	14.8	15.9	19.6	22.0	24.7	27.2	28.0	26.6	23.7	19.1	15.7
		MCDB	14.7	17.5	18.1	21.8	24.5	28.7	31.3	31.8	30.1	26.1	21.4	18.1
	2%	WB	9.2	11.2	14.4	17.9	20.7	23.7	26.6	27.2	25.8	22.3	17.1	12.4
		MCDB	12.2	14.2	17.5	20.8	23.7	26.9	30.3	30.9	28.9	25.1	19.3	15.2
	5%	WB	7.3	9.0	13.0	16.6	19.9	22.9	26.1	26.7	25.2	21.1	15.9	10.6
		MCDB	10.4	12.0	16.2	19.7	22.9	25.8	29.6	30.2	28.3	23.5	18.2	13.3
	10%	WB	6.1	7.2	11.2	15.5	19.0	22.2	25.5	26.3	24.7	19.7	14.8	9.3
		MCDB	9.3	9.9	14.6	18.6	22.1	24.8	29.0	29.7	27.7	22.4	17.3	12.5
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.4	6.8	6.8	6.3	5.4	5.5	5.7	5.0	5.0	5.6	6.1
		MCDBR	8.4	8.7	9.0	8.4	7.7	7.0	6.9	6.8	6.3	6.6	7.1	8.2
		MCWBR	6.5	7.1	6.6	5.3	4.2	3.5	2.9	2.7	3.1	4.6	5.6	6.4
	5% WB	MCDBR	7.6	8.3	8.3	7.1	6.7	6.0	6.3	6.3	5.8	5.9	6.4	7.3
		MCWBR	6.7	7.4	7.0	5.3	4.6	3.3	2.9	2.8	3.2	4.7	5.8	6.6
Clear-Sky Solar Irradiance	taub	0.383	0.432	0.499	0.549	0.570	0.602	0.594	0.581	0.517	0.488	0.430	0.385	
	taud	2.237	2.094	1.936	1.860	1.865	1.857	1.922	1.944	2.090	2.096	2.171	2.250	
	Ebn at Noon	792	790	768	753	743	718	720	719	746	725	730	761	
	Edh at Noon	108	139	177	201	203	204	190	183	151	138	115	101	
All-Sky Solar Radiation	RadAvg	2.79	3.27	4.08	4.77	5.12	4.50	4.95	4.99	3.85	3.02	2.51	2.43	
	RadStd	0.20	0.34	0.26	0.48	0.54	0.43	0.84	0.46	0.35	0.32	0.24	0.19	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (38 neighbors)		(m)								
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page